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Multi-segment trunk kinematics during locomotion and elementary exercises

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ABSTRACT

Background: Motion of human trunk segments in healthy subjects during activities of daily living has been described either with oversimplified models or with cumbersome techniques of isolated anatomical complex. This study describes multi-segmental trunk motion based on a new technique which is a compromise between technical limitations, implied with the experiments, and clinical relevance.
Methods: The thorax segment was tracked by the optimal spatial matching of four thoracic markers. The separate bi-dimensional shoulder line rotations and translations with respect to the thorax were calculated by markers on the two acromions. Spine motion was characterised by a 5-link-segment model from additional four skin markers, in the anatomical reference frame based on four pelvic spine markers. These 14 markers were tracked in 10 healthy subjects and one clinical case during static upright posture, chair rising–sitting, step up-and-down and level walking, and also during elementary flexion and extension, lateral bending, and axial rotation movements of the entire trunk.
Findings: Intra-subject repeatability over ten repetitions was found to be high for most of the measurements, with average standard deviations of less than 1.8° for all planar rotations at the spine, and less smaller than 1 mm for shoulder translations. Large motion, albeit with different patterns, was found in all subjects, also revealing interesting couplings over the three anatomical planes.
Interpretation: Considerable subject-specific motion occurs at each of these different trunk segments in all three anatomical planes, in simple exercises and in motor tasks of daily living. Measurements taken with the present new trunk model in pathological subjects shall reveal corresponding patterns and ranges of motion in abnormal conditions.

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1. Introduction

In clinical movement analysis by stereophotogrammetry the fundamental role of the lumbar and thoracic segments of the trunk has been recognised not only in reaching and upper limb analyses, but also for the functional assessment of locomotor activities, i.e. level walking, step up/down, turning and cutting manoeuvres, stair ascending/descending and chair rising-sitting (Houck et al., 2006; Konz et al., 2006a; Krebs et al., 1992; Leteneur et al., 2009; Skinner and Effeney, 1985; Su et al., 1998). Oscillatory trunk movements and elementary exercises are also used to assess range and quality of motion. The trunk anatomical complex was demonstrated to contribute considerably to achieving and controlling locomotion (Cappozzo, 1983; Kavanagh et al., 2006; Perry, 1992) particularly in regulating oscillations in the three anatomical directions (Kavanagh et al., 2006; Wu et al., 2004). Any neuro-musculo-skeletal pathology, at the trunk as well as at the lower limb, implies deformation and/or impairment and would therefore increase the demand for motion control, which is achieved also by compensatory mechanisms at the upper body (Cappozzo, 1983; Nadeau et al., 2003). The trunk is involved to a large extent in many activities of the daily living (Gebruers et al., 2010) and analysis of movements other than gait has been recently recommended also within the top eight research priorities in clinical movement analysis (Engsberg et al., 2009). Quantitative assessment of trunk kinematics has been sought recently also in sports medicine (Stuelcken et al., 2010).

Trunk motions have been tracked and described using several different models. Full three-dimensional (3D) rotations between a few rigid body segments have been reported (Baker, 2006; Fantozzi et al., 2003; Krebs et al., 1992; Nguyen and Baker, 2004), as well as simpler two-dimensional (2D) projection angles (Bartonek et al., 2002; Davis, 2005; Ferrarin et al., 2002, 2004; Frigo et al., 2003). A number have been proposed to address specific clinical problems (Bartonek et al., 2002; Ferrarin et al., 2002; Gupta et al., 2005; Kramers-de Quervain et al., 2004; Romkes et al., 2007). Most of these models adopt single skin markers, although in very different marker sets, but also rigid clusters have been utilised (Fantozzi et al., 2003; Houck et al., 2006; Konz et al., 2006b; Krebs et al., 1992; Wu et al., 2004). This has produced a variety of technical and anatomical coordinate systems and joint conventions, resulting in confusing

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Nomenclature

AR	axial rotation
C7	spinous processes of the seventh cervical vertebra
ELM-FE	elementary flexion and extension of the trunk
ELM-LB,	elementary lateral bending
ELM-ROT	elementary axial rotations
FE	flexion/extension
IJ	deepest point of incisura jugularis
L1, L3, L5	first, third and fifth lumbar vertebrae
Lab	global laboratory reference frame
LA	left acromion
LASIS	left anterior superior iliac spine
LB	lateral bending
LPSIS	left posterior superior iliac spine
MAI	midpoint between the inferior angles of most caudal points of the two scapulae
Pel	anatomical reference frame of the pelvis
PX	xiphoid process, i.e. most caudal point of the sternum
RA	right acromion
RASIS	right anterior superior iliac spine
RPSIS	right posterior superior iliac spine
Sh	shoulder line segment
Sp1-Sp2-..Sp5	spine line segments
T2	second thoracic vertebrae
Tho	anatomical reference frame of the thorax

terminology. The main difference, however, concerns the individual segments analysed: thorax or the entire trunk all together (Chung et al., 2010; Davis, 2005; Fantozzi et al., 2003; Ferrarin et al., 2002; Gilleard et al., 2008; Kramers-de Quervain et al., 2004; Nguyen and Baker, 2004; Romkes et al., 2007; Su et al., 1998), spine and shoulder line (Frigo et al., 2003), and upper and lower trunk and lumbar spine (Crosbie et al., 1997). The limits and merits of these techniques depend obviously on the relevant physiological or clinical interest. For example, when markers are on the shoulders, their isolated motion can be either the target measure or an artefactual phenomenon, depending on the trunk motion representation. Overall, those models based on marker clusters covering a large area in the frontal plane (Davis, 2005; Fantozzi et al., 2003) are more suitable to describe axial deformities and malrotations in the transverse plane. On the other hand, those models based on anterior thorax and spine markers only (Ferrarin et al., 2002, 2004; Frigo et al., 2003; Nguyen and Baker, 2004 the 'direct' model; Baker, 2006; Wu et al., 2005), are more suitable to describe postural static and dynamic defects in the sagittal plane. A possible combination of these measures can analyse full 3D motion at a number of different segments of this complex body part.

Unfortunately, scarce specific attention has been paid to the physiological motion at this complex with large resolution, i.e. the number of separated segments analysed, and to the reliability of the relevant technique. A recent paper by the present authors has also revealed how wide the spectrum of measurement resulting from the same motion exercise can be when analysed by some of these models (Leardini et al., 2009). In spite of these differences, trunk kinematics data are shared, exchanged and interpreted irrespectively of the model and convention adopted, although it has been recommended that more care toward data sharing and concept understanding should be a top priority in clinical movement analysis (Engsberg et al., 2009).

The present study aims at assessing the amount and the consistency of natural motion at the trunk anatomical complex during locomotion and elementary oscillatory exercises by analysing this at a larger number of segments within the spine, thorax and shoulders, each assumed however to be rigid, both in 2D and 3D. This should support the comprehension and interpretation of the relevant physiological motion, and be the base for future clinical analyses in impaired patients. Because a model for a reliable measure of full trunk motion at a number of separated segments has not yet been established in clinical gait analysis, a new proposal is formulated here for this purpose, as a compromise between practical issues and physiological relevance.

2. Methods

2.1. The marker set and the model

The marker-set included 14 markers (Fig. 1), 4 on the pelvis and 10 on the trunk. These were spherical markers, 14 mm in diameter, glued directly onto the skin with bi-adhesive tape, and placed on the following landmarks (Van Sint Jan, 2007; Wu et al., 2005): the right and left anterior superior iliac spines (RASIS, and LASIS), the right and left posterior superior iliac spines (RPSIS, and LPSIS), the deepest point of incisura jugularis (IJ), i.e. the suprasternal notch, the xiphoid process, i.e. the most caudal point of the sternum (PX), the right and left acromions (RA, and LA), the midpoint between the inferior angles of most caudal points of the two scapulae (MAI), and the spinous

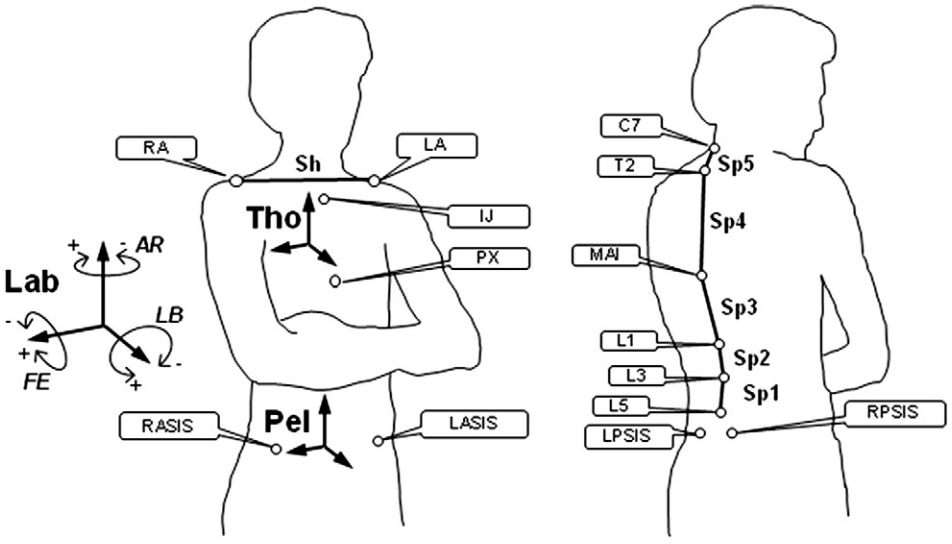


Fig. 1. Diagram with names and locations of the markers used, and views from the front and from the back.

processes of the seventh cervical vertebra (C7), of the second thoracic vertebrae (T2), and of the first, third and fifth lumbar vertebra (L1, L3, and L5).

At the pelvis, the anatomical reference frame (Pel) has the medio-lateral axis between the RASIS and LASIS (positive to the right), the antero-posterior axis as the vector lying in the plane defined by these two landmarks and the mid-point between the PSISs, and orthogonal to the medio-lateral axis (to the front). At the thorax, a technical reference frame was first defined by applying the Single Value Decomposition procedure (Söderkvist and Wedin, 1993) to the T2, MAI, PX and IJ markers for optimal pose, i.e. position and orientation, and estimation. The reference pose, with the centre of the frame in the markers centroid, was obtained by averaging 300 samples from the static upright posture, where the local coordinates of the markers were calculated. At each sample of the motion task the optimal pose was calculated again, along with the global coordinates of the markers. These optimised global coordinates were used to define the thorax anatomical reference frame (Tho), with a sagittal plane defined by PX, MAI and T2, which defines the orthogonal medio-lateral axis (to the right), and a proximo-distal axis between MAI and T2 (upward). Both the relative 3D rotations between the pelvic and the thoracic frames (Tho/Pel) and the absolute 3D rotations between the thoracic and the global laboratory frame (Tho/Lab) were calculated in accordance with the standard joint convention (Grood and Suntay, 1983), thus obtaining flexion/extension (FE), lateral bending (LB) and axial rotation (AR) in the sagittal, frontal and transverse planes respectively. These were deemed to have positive signs when the rotation of the upper segment with respect to either the Lab or the lower segment was toward the front, the right and the right side respectively.

A shoulder line segment (Sh) was defined between RA and LA, and between all adjacent markers on the spine (Sp1-Sp2-..Sp5). Relevant planar rotations were calculated by corresponding projections on relevant anatomical planes, i.e. Sh in the frontal and transverse planes of Tho (Sh/Tho), Sp 1-2-3-4-5 in the sagittal and frontal planes of Pel. For the latter, the angle analysed was the acute between the lines of each two adjacent segments; for Sp1, the angle was between this and the vertical axis of Pel. RA and LA marker translations were also calculated in the Tho frame with origin in T2 (RA/Tho and LA/Tho respectively). All calculations were performed by means of custom-made software programmes implemented in Matlab (The MathWorks, Inc.).

2.2. Subjects, motion tasks and exercises

One volunteer was analysed for intra-subject repeatability (subject #1 in Table 1). Nine additional volunteers were analysed for inter-subject variability. These young subjects had consistent anthropometric characteristics, with no musculoskeletal or neurological diseases. They were asked to perform separately but in the same order, at a self selected speed, the following exercises: static upright posture, armless chair rising–sitting (chair), step up-and-down (step) and level walking (gait). In the former two, the subjects were asked to cross their arms over their abdomen for better marker visibility. The height of the chair and the initial position for the subject were adapted according to the subject's height, so that the tibia was perpendicular to the floor and flexion angles at the hip and the knee joints were about 90°. The feet were parallel and equidistant from the chair (Krebs et al., 1992). Subjects stepped up a 16-cm high, 40-cm wide and 28-cm long step with the right foot first and then with the left foot, and they descended the step with the same foot order. For gait, the cycles of the right leg were analysed. The subjects were then asked to perform elementary, i.e. as planar as possible, movements of the entire trunk with arms crossed, separately over three anatomical planes: flexion and extension (ELM-FE), lateral bending on both sides (ELM-LB), and axial rotations toward both directions (ELM-ROT).

2.3. Data collection and processing

Marker motion was recorded with an eight-camera motion capture system (Vicon 612, Vicon Motion Systems Ltd, Oxford, UK), at a sampling rate of 100 Hz. To label the markers, reconstruct, and in case interpolate the 3D trajectories, Vicon standard software tools

Table 1
Top: anthropometric data of the subjects analysed. Bottom: for each of these subjects the values of all analysed variables are reported for the up-right posture. In addition to the nomenclature as in the text, translations (Transl) up/down or anterior/posterior are reported. Relevant means and standard deviations (SD) are reported in the last two columns.

Anthropometric data and static mean values		Subjects										Mean	SD
		#1	#2	#3	#4	#5	#6	#7	#8	#9	#10		
Gender [male–female]		M	M	M	F	F	F	M	F	M	F	–	–
Age [yy]		25	23	25	25	26	25	25	25	24	24	24.7	0.8
Height [cm]		173	179	185	165	159	164	180	170	173	168	171.6	8.1
Weight [kg]		63	72	74	57	46	57	70	52	70	63	62.4	9.3
BMI [kg/m^2]		21.0	22.5	21.6	20.9	18.2	21.2	21.6	18.0	23.4	22.3	21.1	1.7
Tho/Lab	FE [Deg]	−1.1	7.8	15.8	5.7	2.7	8.4	6.6	2.4	8.7	3.3	6.0	4.6
	LB [Deg]	1.3	−0.8	1.1	0.5	1.7	−1.5	0.8	−1.1	1.8	2.5	0.6	1.3
	AR [Deg]	3.0	−0.1	−1.6	−0.2	0.8	0.1	−0.1	−0.1	−0.1	2.9	0.5	1.4
Tho/Pel	FE [Deg]	−14.6	7.7	3.6	−6.1	−4.8	0.0	−0.6	−10.8	2.4	−12.5	−3.6	7.4
	LB [Deg]	1.5	2.3	−0.1	−3.3	1.8	0.1	3.0	−1.7	0.5	3.0	0.7	2.0
	AR [Deg]	1.9	−0.8	−1.5	1.8	1.8	0.4	−1.5	−1.1	−1.1	1.8	0.2	1.5
Sh/Tho	LB [Deg]	3.7	0.0	−0.8	0.5	1.4	1.3	0.8	0.3	−2.0	−1.7	0.4	1.7
	AR [Deg]	−2.4	0.5	−0.5	−4.8	−3.0	−2.2	0.0	−0.2	−1.2	0.7	−1.3	1.7
RA/Tho	Transl up/down [mm]	0.5	0.2	0.3	0.3	0.2	0.4	0.0	0.0	0.6	0.1	0.3	0.2
	Transl ant/post [mm]	1.0	1.0	0.9	1.0	1.0	1.0	1.2	0.8	0.9	0.8	1.0	0.1
LA/Tho	Transl up/down [mm]	0.7	0.2	0.3	0.4	0.3	0.4	0.0	0.0	0.5	0.0	0.3	0.2
	Transl ant/post [mm]	0.8	1.1	0.8	0.8	0.9	0.9	1.2	0.8	0.8	0.9	0.9	0.1
Sp5/Sp4	FE [Deg]	17.9	14.8	17.2	19.8	19.8	13.4	20.3	26.6	9.8	24.8	18.4	5.0
Sp4/Sp3	FE [Deg]	10.9	25.4	18.6	19.9	15.2	19.8	15.6	22.3	21.4	16.6	18.6	4.1
Sp3/Sp2	FE [Deg]	−0.7	−6.7	7.0	−5.3	−11.5	−0.4	−13.1	−13.3	−4.7	−15.7	−6.4	7.1
Sp2/Sp1	FE [Deg]	−13.3	−20.9	−6.0	−22.9	−13.2	−27.1	−11.4	−22.6	−10.1	−13.7	−16.1	6.8
Sp1/Pel	FE [Deg]	−11.5	9.9	−16.0	2.1	4.6	8.0	8.2	2.8	−4.1	0.3	0.4	8.6
Sp5/Sp4	LB [Deg]	4.3	−5.8	−7.0	0.8	−0.9	1.6	0.7	1.0	1.8	−2.1	−0.6	3.5
Sp4/Sp3	LB [Deg]	3.8	2.7	2.8	2.8	−0.4	−1.5	3.7	−3.5	4.2	5.8	2.1	2.9
Sp3/Sp2	LB [Deg]	−0.7	0.0	−1.4	−3.8	4.2	9.8	−3.2	1.3	0.0	0.4	0.7	3.9
Sp2/Sp1	LB [Deg]	−1.9	−3.3	−1.5	0.7	−2.6	−5.6	2.1	7.4	−2.6	−1.5	−0.9	3.6
Sp1/Pel	LB [Deg]	0.4	3.1	0.0	−3.1	0.6	−2.7	0.4	−7.0	−1.2	−1.6	−1.1	2.7

Table 2
Trunk motion at the joints and segments analysed; AvgSD and AvgRoM (see text for their meaning) over 10 repetitions of subject #1.

Intra-subject repeatability		ELM-FE		ELM-LB		ELM-ROT		Chair		Step		Gait	
		Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM
Tho/Lab	FE [Deg]	4.1	51.9	1.0	6.0	1.0	3.1	4.5	46.8	2.2	12.9	0.8	2.6
	LB [Deg]	0.8	3.0	2.8	52.6	0.7	2.4	1.5	5.1	1.4	7.2	0.8	4.7
	AR [Deg]	1.2	2.0	2.3	10.2	4.1	52.5	2.4	6.8	2.1	10.7	1.0	5.9
Tho/Pel	FE [Deg]	3.1	33.6	1.0	5.2	1.0	3.2	4.2	43.1	1.9	10.4	0.8	3.4
	LB [Deg]	0.9	2.1	2.6	47.3	0.8	2.5	1.3	5.3	1.8	15.7	0.9	13.9
	AR [Deg]	0.5	2.7	1.4	19.5	2.2	37.3	1.5	5.9	1.1	9.5	0.9	14.5
Sh/Tho	LB [Deg]	0.6	1.0	0.8	6.2	0.7	2.3	0.6	2.2	0.6	1.3	0.4	3.3
	AR [Deg]	0.4	0.9	0.5	2.6	1.5	16.6	1.0	2.1	0.5	1.2	0.6	3.4
RA/Tho	Transl up/down [mm]	0.3	1.1	0.5	1.8	0.3	0.9	0.8	2.4	0.3	0.5	0.3	1.1
	Transl ant/post [mm]	0.3	0.9	0.7	1.0	0.6	4.8	0.7	1.8	0.4	0.5	0.5	1.2
LA/Tho	Transl up/down [mm]	0.3	0.8	0.4	1.7	0.3	0.7	0.7	2.0	0.3	0.4	0.2	1.0
	Transl ant/post [mm]	0.3	0.8	0.6	0.7	0.6	4.7	0.9	1.6	0.5	0.6	0.4	1.0
Sp5/Sp4	FE [Deg]	0.5	1.6	0.7	3.9	0.8	5.8	1.3	6.7	0.4	2.0	0.7	3.1
Sp4/Sp3	FE [Deg]	1.0	3.6	0.7	2.9	0.6	3.1	1.8	6.6	0.6	2.0	0.5	1.3
Sp3/Sp2	FE [Deg]	1.1	7.7	0.6	2.9	0.8	5.4	1.4	7.9	0.7	1.8	0.4	1.9
Sp2/Sp1	FE [Deg]	1.3	17.0	0.9	3.7	1.2	6.3	1.5	17.6	1.2	6.7	0.7	3.5
Sp1/Pel	FE [Deg]	1.2	11.2	0.6	2.7	0.8	3.3	1.6	26.4	0.8	5.3	0.7	2.9
Sp5/Sp4	LB [Deg]	0.8	2.7	1.2	13.1	1.6	18.0	1.7	4.4	0.8	3.0	0.8	9.9
Sp4/Sp3	LB [Deg]	0.4	1.8	1.3	18.3	0.8	4.8	0.9	2.4	0.6	2.2	0.4	6.9
Sp3/Sp2	LB [Deg]	0.4	1.3	1.1	15.4	0.8	8.7	0.6	2.7	0.6	3.0	0.5	3.4
Sp2/Sp1	LB [Deg]	0.3	2.4	1.0	8.4	0.9	5.3	0.5	2.6	0.8	6.3	0.7	5.6
Sp1/Pel	LB [Deg]	0.6	2.0	0.6	3.6	0.8	3.5	1.0	4.3	0.9	8.0	0.4	5.5

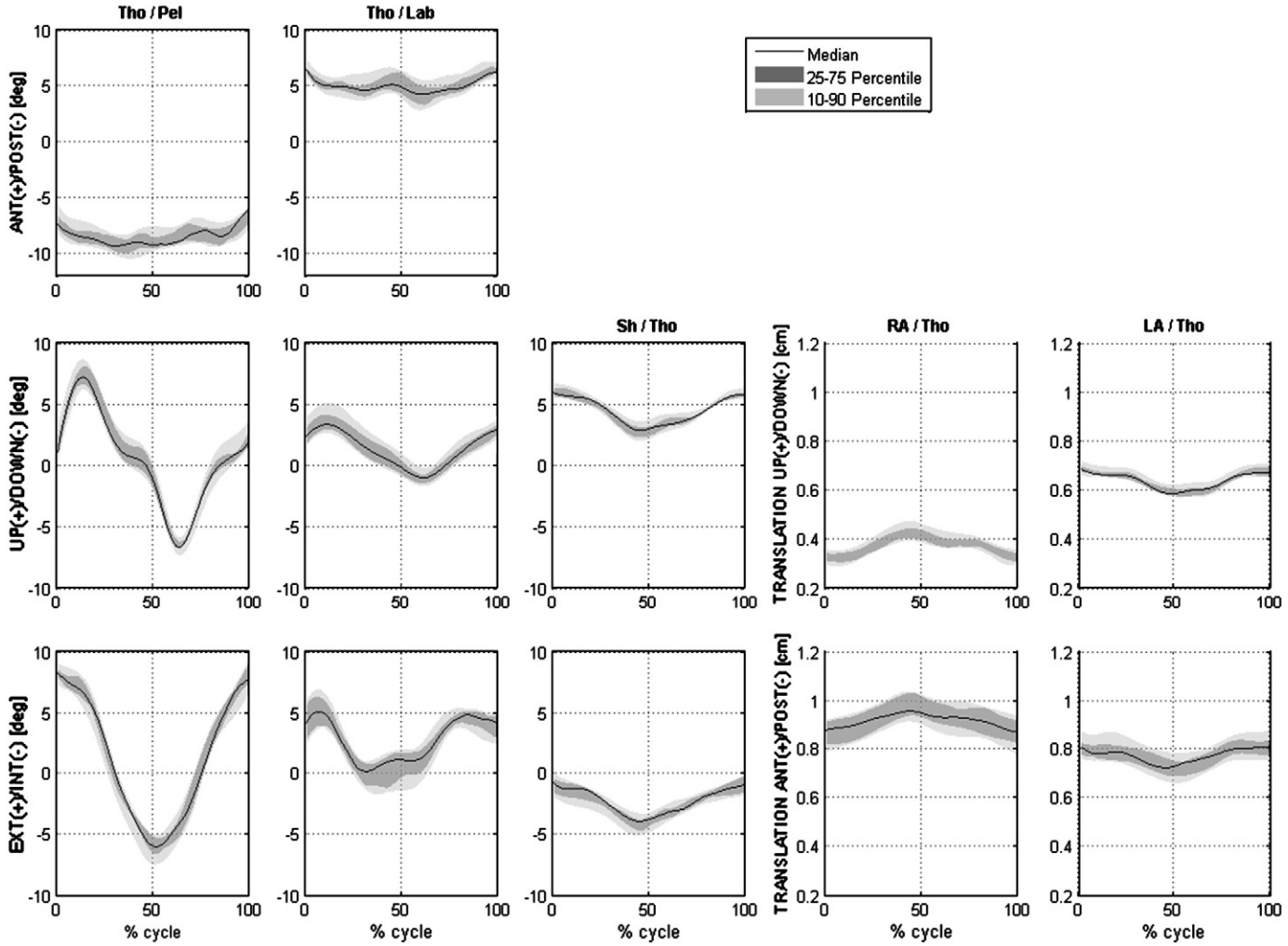


Fig. 2. Main trunk motion patterns in the sagittal (top row plots), frontal (central row) and transverse (bottom row) planes, represented as median and percentiles over ten repetitions of the right gait cycle for subject #1. Rotations (first three column plots) and translations are reported (last two column plots). For the rotations, positive was defined as flexion towards the anterior side of the body (ANT), bending to the right side (UP) and axial rotation to the right side (EXT); negative was the opposite motion, respectively POST, DOWN, INT. The median curve (solid dark line), the 25–75th percentile band (dark grey) and the 10–90th percentile band (light grey) are shown.

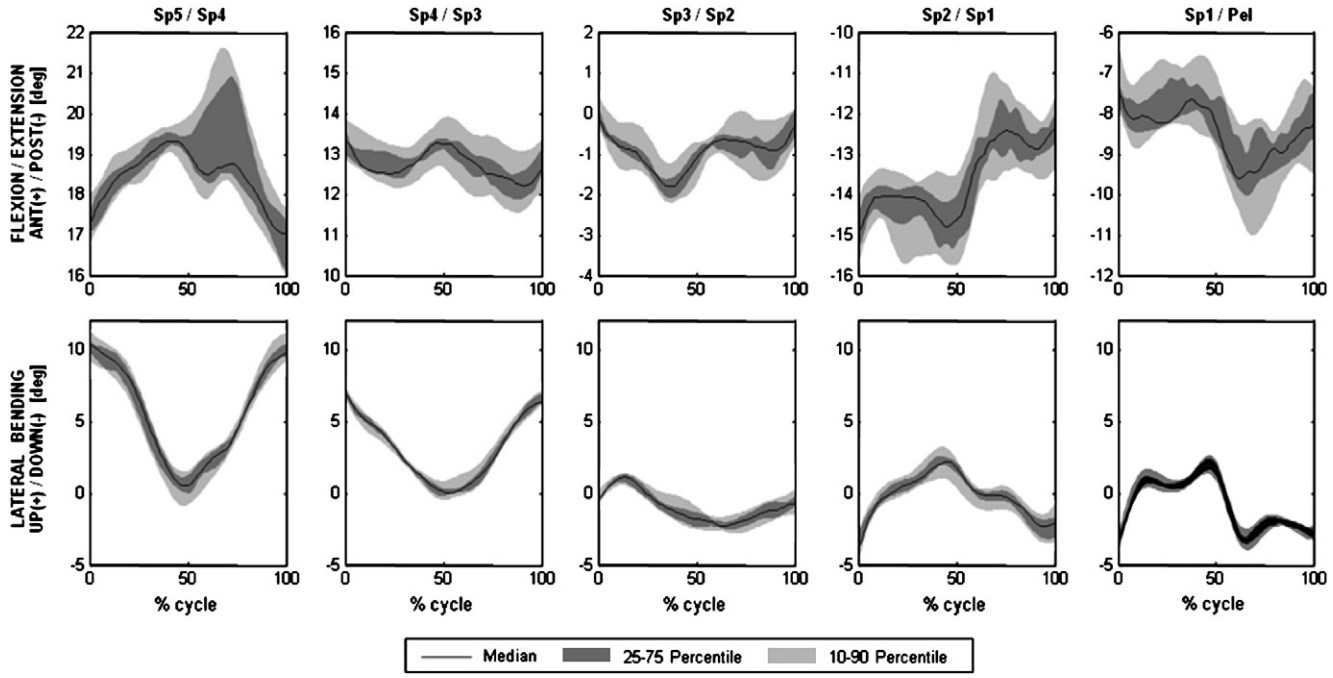


Fig. 3. Angle patterns between line segments at the spine, in the sagittal (top plots) and frontal (bottom) planes, represented as median and percentiles over ten repetitions of the right gait cycle for subject #1. The median curve (solid dark line), the 25–75th percentile band (dark grey) and the 10–90th percentile band (light grey) are shown.

were used. In particular, marker trajectories were smoothed using Woltring's smoothing splines (Mean Square Error = 10) (Woltring, 1986), implemented in these tools. Cycle identification was obtained by a single experienced operator on standard procedures based on force plates and foot marker trajectories in gait. In all other tasks, the cycles were manually identified over standard stick diagram animations, by looking at the beginning and end of the relevant trunk marker movement. These selected data intervals were resampled on a 0–100 basis to allow for curve comparison. For each task and variable, the average of the range of motion (AvgRoM) was calculated. In addition, intra-subject repeatability was assessed by the standard deviation at each sample of the curves obtained from the 10 repetitions performed by the subject, averaged over the cycle (AvgSD). Similarly, for inter-subject variability, the standard deviation at each sample of the curves obtained from one repetition from each of the 10 subjects was averaged over the cycle.

All motion patterns, both for intra-subject repeatability and inter-subject variability, were reported by a novel graphical representation. At each normalised sample within the gait cycle, the 10th, 25th, 50th (median), 75th and 90th percentile were computed, represented by a central median curve, and the 10–90th and 25–75th percentile bands.

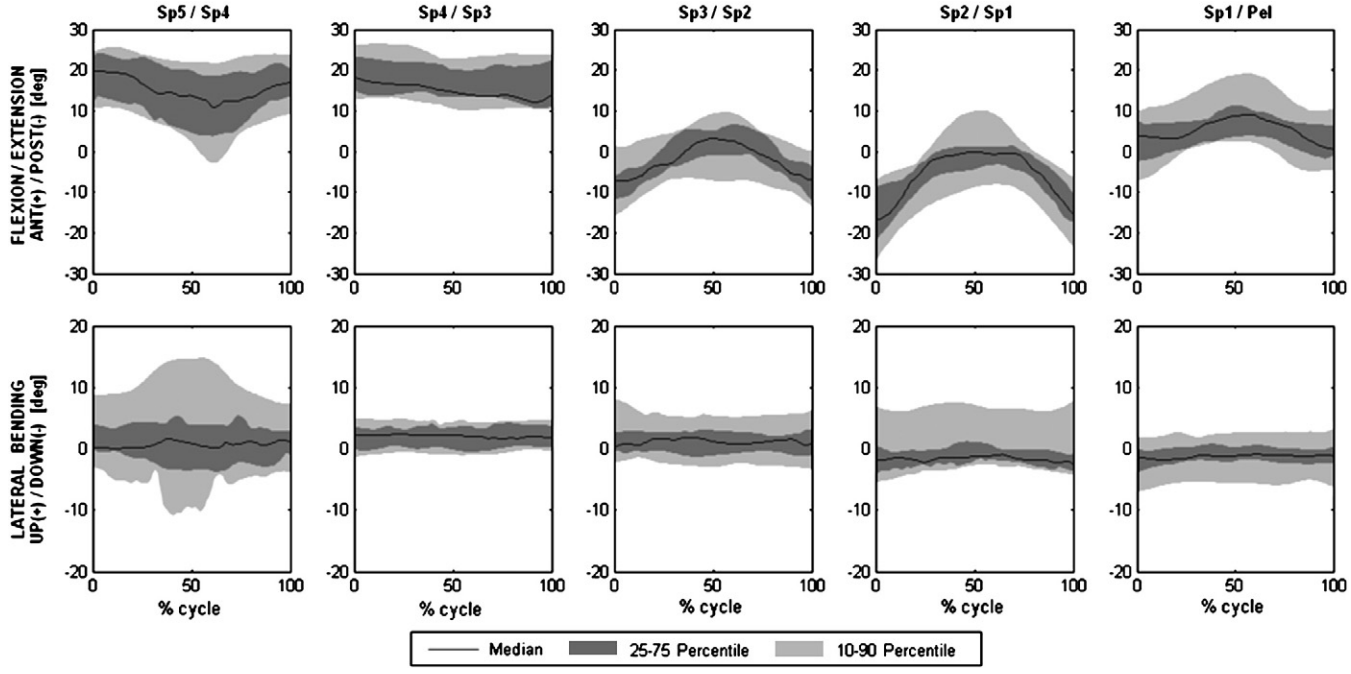


Fig. 4. Angle patterns between line segments at the spine, over the ten subjects for the ELM-FE exercise (same graphical representation and sign convention as for Fig. 3).

3. Results

Intra-subject repeatability was high for most of the measurements (Table 2). AvgSD was less than 1.8° for all planar rotations at the spine, and less than 1 mm for all translations. As expected, this was greater in the elementary exercises with no specific tasks to perform, but still all within 4.1° (Tho/Lab in ELM-FE and ELM-ROT), which corresponded however to an AvgRoM of about 52°. The more constraining the exercise was, i.e. with a target to achieve, the more repeatable the rotations were, Tho/Lab rotations being all less than 4.5° in chair, 2.2 in step, and 1.0 in gait (see also Figs. 2 and 3), for an AvgRoM of about 47, 13 and 6° respectively. Consistent patterns were also observed for small motion (Figs. 2 and 3), even for the 1÷2 mm in RA–LA/Tho translations, the 3÷4° of Sh/Tho and spine planar rotations. Joint motion not reported before in the literature was, in this exemplary subject, particularly consistent and had a considerable AvgRoM: Sh/Tho rotation which reached more than 17° in ELM-ROT, 6 in ELM-LB, 3° in gait; RA and LA translations reached 5 mm in ELM-ROT and 2 mm in chair; lumbar spine (Sp1, and Sp2) planar rotations reached 18 and 26° in chair, about 3° even in gait; cervical spine (Sp5) reached nearly 7° in chair, and 10° in gait.

Large motion was found in all segments and joints, and in all subjects analysed, although the patterns were consistent for some variables in certain motor tasks and different from subject-to-subject in others (Fig. 4 and Table 3; a full series of plots for the patterns of inter-subject variability is shown in the Appendix A). For example, FE of Tho/Pel and Tho/Lab showed a larger variability, thus indicating differences amongst subjects in the strategies of trunk flexion. Shoulder translations were not large for each subject, although the direction of this motion was consistent. The large variability of a number of variables during motion might be partially due to the large differences of the corresponding values in upright posture (Table 1), in particular thorax flexion and spine segment rotations. These values can be removed from each corresponding variable during motion, i.e. the common off-set issue, but this is rarely performed in trunk kinematics analysis.

As an exemplary clinical application, the patterns of motion during Chair rising from a left hemiplegic patient after a stroke are superimposed on the corresponding bands from the present healthy subjects (Fig. 5). This reveals an abnormal orientation of the thorax (Tho/Lab), with premature flexion, an initial attitude in right bending partially corrected during rising, and an initial attitude also in rotation to the right; these are observed at the Tho/Pel relative motion to a much lesser extent, suggesting spatial orientations at the pelvis. The shoulder line (Sh/Tho) is also inclined down to the left side with respect to the thorax, as for the upper position of the healthy right acromion (RA/Tho) at the beginning of the task. Its posterior translation results in the rotation to the right of the shoulder line.

4. Discussion

The general aim of this work was to describe human trunk kinematics during locomotion and elementary exercises with a larger overall resolution than that reported in the literature so far. An original clinical-oriented model for these measurements was devised, and relevant feasibility and intra-subject repeatability assessed. Spine, thorax and shoulders have been analysed largely in the past, and each anatomical complex perhaps in more detail than the present study, but this has been done in separate studies, with contrasting results, and rarely including locomotion activities. The extent to which these various parts of the trunk contribute to the necessary oscillations of the upper body in the three anatomical planes for the control of balance and locomotion, for example, has not yet been established. Little is known about these contributions even in elementary exercises of the upper body, often analysed in a clinical setting to assess the degree of trunk deformity and mobility (Bruijn et al., 2008; Gombatto et al., 2007). Nowadays, these scopes are considered very important in clinical gait analysis (Engsberg et al., 2009; Jaspers et al., 2009; Konz et al., 2006a).

At this part of the body, representation of motion at all the relevant subparts is very complex, because it occurs at many different small joints. Amongst the most complex thorough motion depiction at each single joint and the simplest overall motion of a single rigid segment, intermediate representations should be identified as a compromise between experimental/technical limitations and clinical relevance. The thorax as a whole represents a fundamental part of the trunk, and

Table 3
Trunk motion at the joints and segments analysed; AvgSD and AvgRoM (see text for their meaning) over the 10 subjects analysed.

Inter-subject variability		ELM-FE		ELM-LB		ELM-ROT		Chair-rising		Chair-sitting		Step		Gait	
		Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM	Avg SD	Avg RoM
Tho/Lab	FE [Deg]	8.9	62.3	7.5	8.1	4.0	9.3	7.8	37.8	6.7	39.0	5.2	10.3	4.7	4.2
	LB [Deg]	2.2	3.0	6.6	74.4	4.9	15.0	1.6	2.7	1.6	2.8	2.2	5.6	2.1	5.1
	AR [Deg]	2.6	2.7	6.1	9.6	15.1	125.8	3.3	3.8	3.4	5.4	3.3	12.7	3.1	8.3
Tho/Pel	FE [Deg]	9.9	26.3	9.2	7.0	7.4	6.1	11.8	26.4	10.3	25.4	7.1	10.4	9.0	4.2
	LB [Deg]	2.8	2.9	5.5	61.9	4.3	12.5	2.2	3.5	2.0	3.2	2.7	14.8	2.3	12.3
	AR [Deg]	2.0	3.0	3.9	9.8	5.8	50.0	3.0	3.2	2.5	2.8	2.2	6.2	2.7	11.6
Sh/Tho	LB [Deg]	2.0	1.1	2.6	4.5	2.4	5.4	2.0	1.9	2.2	1.9	1.8	1.6	2.2	2.1
	AR [Deg]	2.1	1.0	2.7	3.2	2.9	15.0	2.5	1.6	2.3	1.2	2.0	1.2	1.8	2.2
RA/Tho	Transl up/down [mm]	2.2	0.5	1.9	1.3	1.8	1.5	2.8	2.0	2.7	1.4	2.1	0.5	2,3	0.7
	Transl ant/post [mm]	1.5	0.8	1.7	1.0	1.5	4.1	1.6	1.3	1.8	1.3	1.6	0.5	1.5	0.7
LA/Tho	Transl up/down [mm]	2.4	0.5	2.3	1.4	2.1	1.5	3.3	1.9	3.3	1.2	2.2	0.6	2.8	0.8
	Transl ant/post [mm]	1.6	1.0	1.5	1.2	1.5	4.1	1.9	1.4	2.2	1.5	1.6	0,6	1.2	0.7
Sp5/Sp4	FE [Deg]	7.4	9.1	5..5	5.4	5.0	6,7	5.6	11.4	4.9	8..4	5,1	3.3	5.4	2.7
Sp4/Sp3	FE [Deg]	5.3	5..1	5.5	3..7	5.1	4.3	4.0	7.3	3.4	5.5	4.5	3.0	4.3	1.6
Sp3/Sp2	FE [Deg]	5.2	12.1	7.3	4.6	7.5	4.7	6.8	9.6	6.0	9.1	6.8	5.4	8.3	3.9
Sp2/Sp1	FE [Deg]	5.9	17.3	7.3	5.5	8.1	5.5	6.9	13.1	6.5	12.2	9.6	6.9	9.4	5.6
Sp1/Pel	FE [Deg]	5.8	8.4	8.5	3.0	9.2	4.2	4.2	14.8	3.7	13.5	7.3	5.1	8.0	3.5
Sp5/Sp4	LB [Deg]	6.5	6.0	3.9	11.1	4.9	23.8	6.0	4.4	6.5	3.9	4.9	4.4	6.4	8.2
Sp4/Sp3	LB [Deg]	2.1	1.7	3.2	17.5	4.3	19.5	2.3	2.7	2.1	2.9	2.5	4.0	2.6	8.3
Sp3/Sp2	LB [Deg]	3.2	3.1	4.9	24.8	5.2	10.8	3.4	3.9	3.2	3.2	3.5	5.5	3.3	4.4
Sp2/Sp1	LB [Deg]	4.0	2.9	4.1	14.1	3.9	5.5	4.4	3.2	4.3	2.6	3.5	5.7	4.2	4.7
Sp1/Pel	LB [Deg]	3.0	2.4	3.1	8.5	3.7	6.2	2.9	3.7	2.8	3.2	2.6	8.3	3.0	6.7

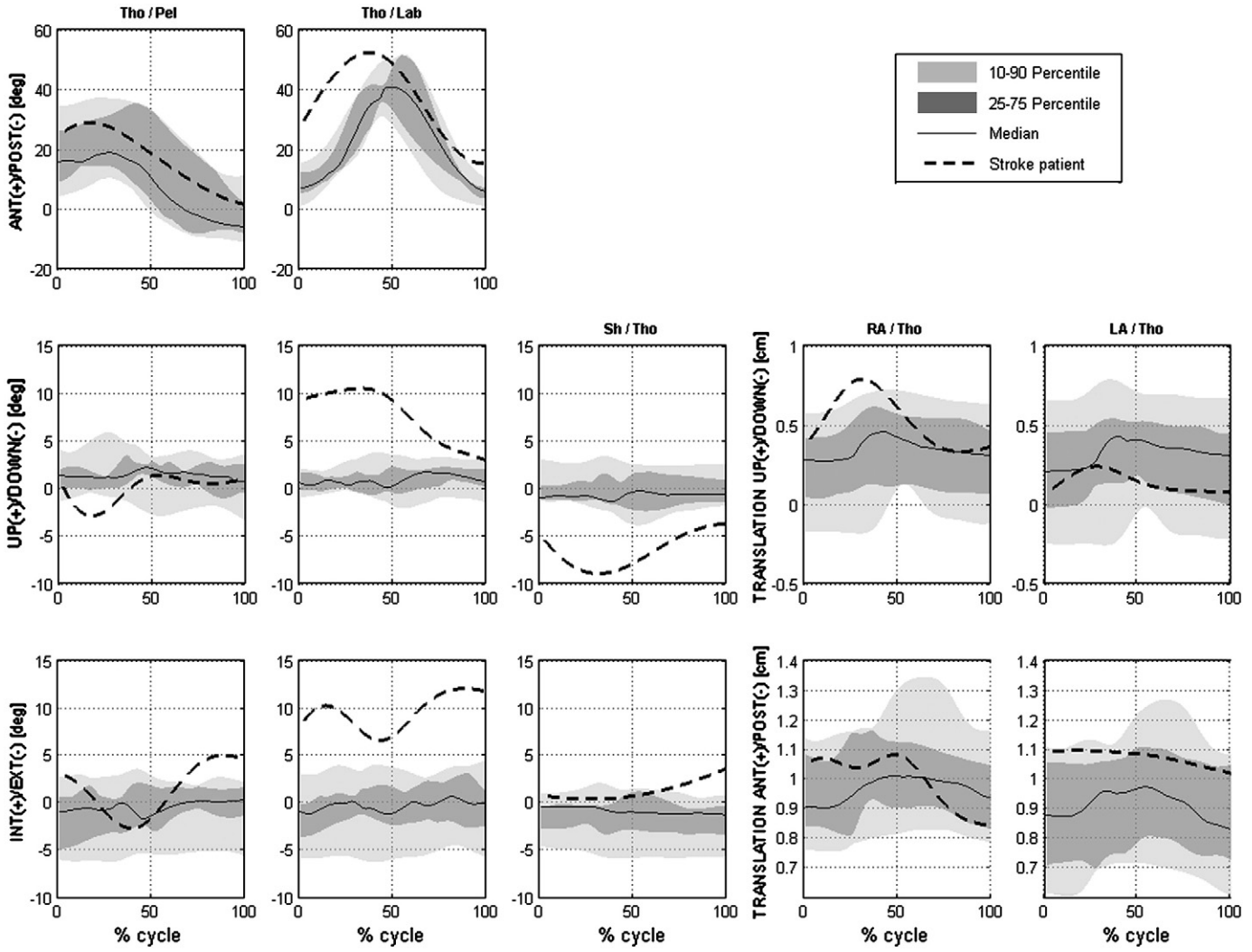


Fig. 5. Main trunk motion patterns over the ten subjects for the chair motor task (same graphical representation and sign convention as for Fig. 2, apart for axial rotation whose positive values are to the right side but termed INT). Corresponding patterns from a patient after stroke (dashed) are superimposed.

except for its deformation in respiratory medicine, a single rigid model would be acceptable in clinical gait analysis. The spine is made up of many vertebrae, not accessible for 3D motion tracking, and somehow too close for the visibility of corresponding reflective markers (Chockalingam et al., 2002; Konz et al., 2006b). The shoulder girdle is complicated and difficult to access, but separate tracking, albeit minimally, seems to be an important step forward with respect to current clinical gait analysis protocols. Therefore, the compromise proposed here includes one shoulder and five spine line segments with 2D motion, and a single 3D rigid body for the thorax, which to the authors' knowledge is the most detailed model in clinical gait analysis. It is expected to be especially useful in orthopaedics (low back pain, scoliosis, thoracic hyperkyphosis, spine surgery, shoulder pathologies, etc.), neurology (trunk control monitoring and effect of therapies in trunk control recovery in Parkinson's disease, and hemiplegic and cerebral palsy patients), rehabilitation (functional diagnosis and outcome measure) and sports medicine (trunk-shoulders motion during specific tasks).

In the past, anatomical frames for the thorax were defined to coincide with relevant technical clusters, based mostly on only three markers. Because the overall axial rotation at the spine is achieved with such small rotations at each vertebra, very different values can be found if this triad has a single vertex at the top or at the bottom of the thorax (Leardini et al., 2009). An optimal pose estimation technique is proposed, which describes relevant motion in its most possible global representation, i.e. by four largely distributed markers over identifiable landmarks. We also focused on the thorax reference frame being based on markers on the thoracic cage and spine, although this does not fully follow ISB recommendations (Wu et al., 2005), which include C7. In front, IJ and PX markers were selected also in previous models (Baker, 2006; Nguyen and Baker, 2004; Wu et al., 2005). As for a fourth marker, clearly in the postero-distal location, T8, T10 and T12 were hypothesised together with the MAI (Wu et al., 2005). Preliminary experience in a number of normal young subjects had revealed difficulties associated with the skin marker location at single thoracic vertebrae and relevant manoeuvres, which resulted in scarce repeatability. The identification of the distal apex of each scapula was much simpler and more repeatable. The scapulae can be asymmetrical and differently elevated in pathological subjects, but because this landmark is one of the four for thorax pose estimation, this is not as critical as it is in other kinematic models. It has been shown however that this landmark usually corresponds to the level of T9 (Haneline et al., 2008). The number and location of markers was also chosen such as to minimise subject disturbance and optimise visibility. Only during static upright posture, chair rising–sitting, and elementary movements, arm crossing over the abdomen was necessary, because of continuous obscuration of a few markers.

The model used in the present study is still limited in terms of resolution. A number of studies have reported on spine rotations at each vertebra, but these models concern explicitly only elementary exercises

in special camera configurations; the present 5-link-segment model may represent a good compromise between high resolution and feasible experiments in clinical gait analysis. Furthermore, the relevant marker-set does not target global geometry for the spine, i.e. kyphosis and lordosis as in Frigo et al. (2003), and therefore cannot represent directly deformation and motion of relevant curves. Conversely, the approach is to target separately cervical, thoracic and lumbar spine segments, compatible with the marker number and visibility. The model does not include the head segment, but this extension is easily achieved by replicating the concepts here applied for the thorax, i.e. four technical markers and an anatomical reference frame. This additional segment represents thoroughly kinematic control in the upper body, and also justifies unusual patterns at the cervical spine, i.e. Sp5. Shoulder girdle motion is not tracked either, mainly because it requires specialised scopes, assumptions, motor tasks and techniques. Skin motion artefact remains an issue for all in-vivo studies based on stereophotogrammetry, and very little has been reported explicitly on this anatomical complex (Heneghan and Balanos, 2010). Recent reports on intervertebral motion (Gercek et al., 2008; Rozumalski et al., 2008) show that the skeletal motion can be even larger than that calculated with skin markers. We expect critical effects only at the spine markers, particularly in those close to each other, in very large spine deforming movements.

The new model for trunk tracking presented in this study demonstrated several advantages. The trunk motion showed intra-subject repeatability and inter-subject variability, which all together implies that subject-specific motion can be tracked with reliability. The present measurements were all taken by a single operator, and the important effect of multiple operators and multiple sessions on the overall repeatability of the technique will be addressed specifically according to established procedures (Manca et al., 2010; Schwartz et al., 2004). Although a consistent pattern of mean motion cannot be obtained from healthy subjects, it is still important in the clinical context to be able to track this motion accurately, in order to quantify the level of impairment or the subject-specific progression of improvement or deterioration under treatments. The present model can overcome one of the major limits of the previously proposed techniques, that is either the inclusion of the shoulder markers in the overall trunk rigid segment or their exclusion; this obviously depends on the clinical or physiological interest of the study utilising trunk motion representation, where shoulder marker motion can be either

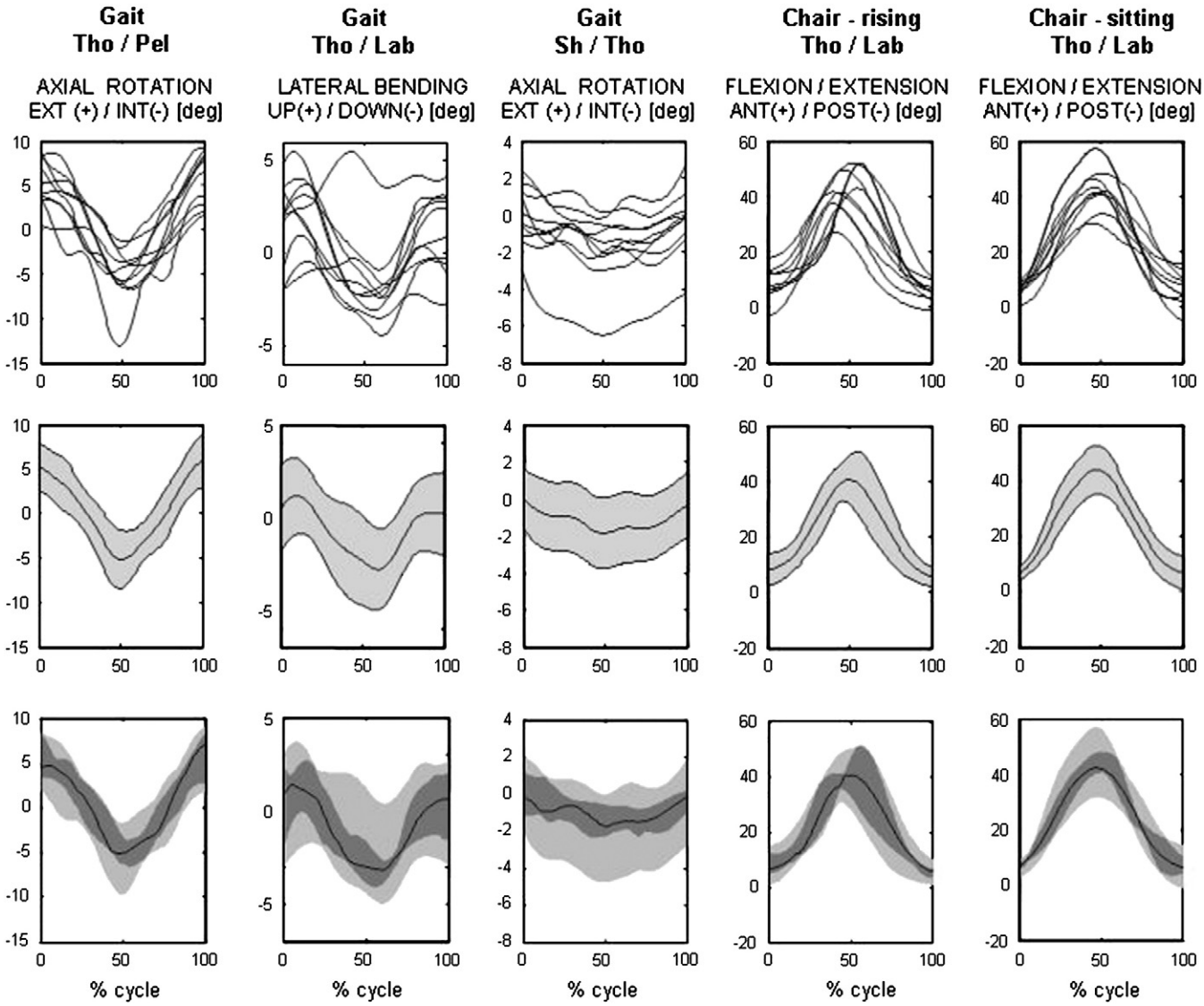


Fig. 6. The novel graphical representation according to the median and percentiles (bottom row plots) for a number of kinematics variables from different motor tasks, together with corresponding patterns obtained with the traditional average and standard deviation (central row), and the corresponding original single curves from the ten subjects analysed (top row). Same sign convention than other figures.

the target or an artefactual phenomenon. This is addressed here by highlighting the isolated motion of the shoulder line, though in 2D. The graphical representation of the typical bands for patient-to-control assessments is also addressed in the present study, tentatively to enhance the typical mean curve and standard deviation bands. This new representation is able to distinguish between two percentile ranges (10–90th and 25–75th), and also represents deviations differently above and below the median curve, both for the magnitude and for the timing of the peaks (Fig. 6). It can also be used on small sample sizes and it is robust to outliers. To the authors' knowledge none of the previous reports from the literature enables appropriate comparisons with the present results, concerning the differences either in the marker-set or in the reference frame definition, or in the motor tasks or the populations analysed.

The present measurements with the relevant technique can also contribute to establish the extent to which the various parts play a role in the overall motion of the trunk (Gombatto et al., 2007; White and Panjabi, 1990). In one subject, the entire FE during the ELM-FE, LB during ELM-LB, and AR during ELM-ROT of the global thorax motion (Tho/Lab) are achieved at the Tho/Pel articulation respectively for the 64.7%, 89.9% and 71.0%, the remaining percentage must be performed by the pelvis. These values were 42%, 83% and 40% on average over the ten subjects. Similar analyses can be performed also for planar rotations of the spine segments; it is worth noting in this respect that thorax 3D flexion and bending can also be explained by summing up corresponding planar rotations at Sp1, Sp2, Sp3, and Sp4. The present model can also characterise better the complex pattern of axial rotation of the trunk segments necessary to secure stability at the head by managing the largely asymmetrical motion at the lower limbs and pelvis (Bruijn et al., 2008); Sh/Tho rotation in the transverse plane (AR) was found as large as 15° in the elementary trunk rotation exercise as averaged over the subjects (Table 3), and reached more than half of the Tho/Lab rotation in gait (3.4 vs 5.9, Table 2).

The present complete 2D or 3D motion representation revealed that there is considerable motion out of the main anatomical plane in elementary exercises, particularly LB in ELM-FE, FE in ELM-LB, but also FE and LB in ELM-ROT (see figures in Appendix A). In the latter, FE at the spine segments was between about 4 and 7°, and LB between about 5 and 24° on average (Table 3). Part of this might be due to artefacts associated with skin motion (Heneghan and Balanos, 2010) and rotation calculations when projection angles are implied, but also subject-specific motor strategies can be deduced. This coupling of motion over the three anatomical planes has been shown to have a physiological component, associated with the geometry of the facet joints (Kozanek et al., 2009; White and Panjabi, 1990).

In the present study motion at a number of trunk segments is reported during locomotion and elementary exercises from a population of healthy subjects. For this purpose, the thorax was tracked from optimal spatial matching of four thoracic markers, a separate shoulder line motion is reported with respect to the thorax, and the spine motion characterised by a 5-link-segment model. In conclusion, the considerable subject-specific motion found at each of these different trunk segments in all three anatomical planes, in simple exercises as well as in motor tasks of daily living such as gait and chair, supports the feasibility of these measures and the relevance of the corresponding measurements in clinical gait analysis studies.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.clinbiomech.2011.01.015.

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